8. Write a program to implement Linear Regression (LR) algorithm in python.

CODE:

```
[8]
✓ 0s
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
# Load Diabetes dataset
data = load_diabetes()
X = data.data
y = data.target
# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1
)
# Create Linear Regression model
model = LinearRegression()
# Train the model
model.fit(X_train, y_train)
# Predict
y_pred = model.predict(X_test)
# Display results
print("Predicted Values:")
print(y_pred[:10])   # First 10 predictions
print("\nActual Values:")
print(y_test[:10])   # First 10 actual values
print("\nMean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
```

OUTPUT:

```
Predicted Values:
[117.35958674 107.58949177 186.61224199  65.02265492 172.78617296
 191.63955511 221.50642356 119.88812724 156.46084971 127.89259776]

Actual Values:
[ 78. 152. 200.  59. 311. 178. 332. 132. 156. 135.]

Mean Squared Error: 2827.084017424082
R2 Score: 0.43845439143447806
```